

Unit 2, Lesson 9: Experimental and Theoretical Probabilities from Simulations

Benchmarks

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Target

- ☐ I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.

2.9.1 Warm-Up: Visualizing Probabilities

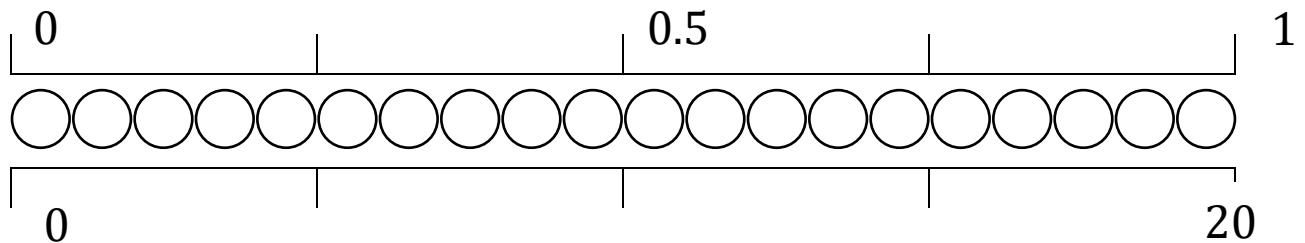
Ashley places 2 red marbles, 3 blue marbles, 4 yellow marbles, 5 orange marbles, and 6 purple marbles in a bag.

A representation of the marbles is shown.

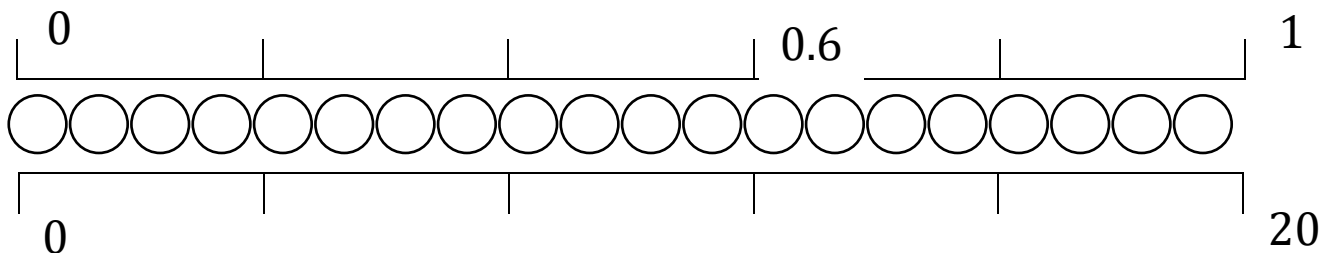


1. Complete the double number line diagrams below by writing values at each tick mark. The top line represents probabilities, and the bottom line represents the number of marbles in the bag.

a.



b.

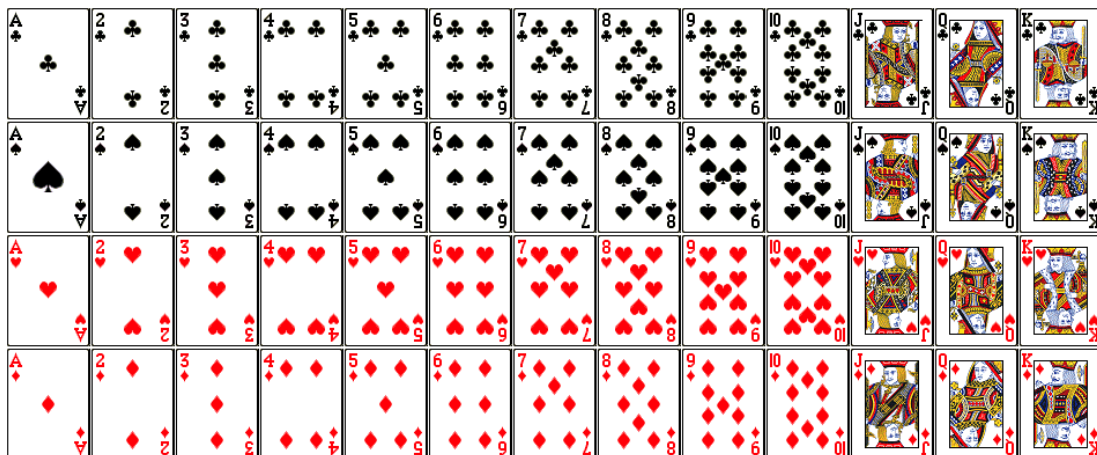


2. Write a color on each blank to complete the statements.

The probability of pulling a _____ marble out of the bag without looking is 0.2. The probability of pulling a _____ marble out of the bag without looking is 0.25.

2.9.2 Collaboration: Simulating Probabilities

1. In the 2019-2020 NBA regular season, Miami Heat player Jimmy Butler made $\frac{1}{4}$ of the three-point shots he attempted. What was the probability of Jimmy Butler making a three-point shot in the 2019-2020 regular season, represented as a decimal?
2. Work with your partner to complete each statement to describe ways Jimmy's three-point shot probability could be simulated using different simple experiments.
 - a. Create a spinner with ____ equal section(s), where ____ section(s) represent(s) Jimmy Butler making a three-point shot and the other(s) represent(s) misses.
 - b. Place twelve equal-sized slips of paper with the numbers 1 – 12 written on them in a bag. Papers with the numbers _____ represent Jimmy Butler making a three-point shot and the others represent misses.
 - c. Use a standard deck of 52 cards where _____ represent Jimmy Butler making a three-point shot and the other cards represent misses.



3. Suppose in the first post-season game that year, Jimmy Butler attempted 5 three-point shots. Choose one of the simulations from the previous question and work with your partner or group to simulate the number of three-point shots Jimmy made.

- a. Repeat the simulation 5 times and record your results in the table.

	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Fraction Made
Simulation 1						$\frac{\quad}{5}$
Simulation 2						$\frac{\quad}{5}$
Simulation 3						$\frac{\quad}{5}$
Simulation 4						$\frac{\quad}{5}$
Simulation 5						$\frac{\quad}{5}$

- b. Based on your group's simulations, estimate the probability that Jimmy Butler made $\frac{2}{5}$ of the three-point shots he attempted in the game.
- c. Jimmy Butler actually made 3 out of the 5 three-point shots he attempted in the game. Explain how that compares to your group's simulation results.
4. In the 2019-2020 NBA regular season, Orlando Magic player Aaron Gordon made $\frac{2}{3}$ of free-throw shots attempted. Work with your partner to describe how each of the following could be used to simulate his attempted free-throw shots.
- a. A fair 6-sided die
- b. A standard 52-card deck
- c. Different colored marbles placed in a bag

2.9.3 Your Turn!

1. Suppose a fair six-sided die numbered 1 through 6 is rolled 12 times.
- a. Of those 12 rolls, how many do you expect will land on the number 3?

_____ times out of 12 rolls
- b. Roll a standard six-sided die 12 times and record the number on the top of the cube when it lands.

Roll 1	Roll 2	Roll 3	Roll 4	Roll 5	Roll 6	Roll 7	Roll 8	Roll 9	Roll 10	Roll 11	Roll 12

- c. What is the theoretical probability of rolling a 3?

d. Describe how the theoretical probability compares to your experimental probability.

e. Review your results and your partner’s results. Explain why the results are not the same.
2. Consider the events of rolling a fair 10-sided die and getting an odd number or flipping a coin and having it land tails-up. Explain which event is more likely.

3. Noelle picks a card at random from a stack that has cards numbered 11 through 25.

a. Determine the sample space.

b. Find each probability.

$$P(\text{even}) =$$

$$P(\text{multiple of 5}) =$$

$$P(\text{two digit number}) =$$

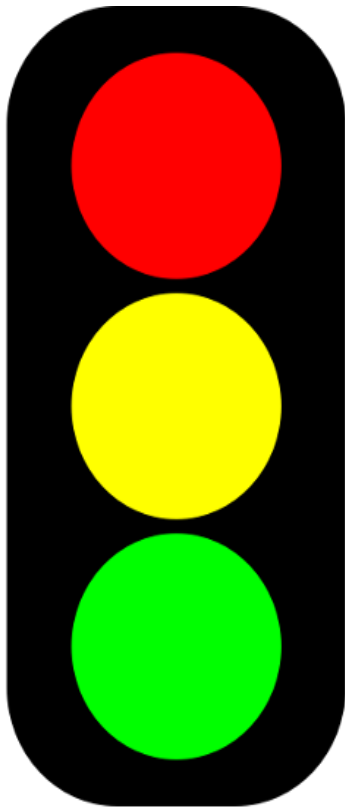
$$P(\text{number less than 10}) =$$

- c. Noelle used a simulator to run the experiment 300 times. She selected an odd-numbered card 154 times. Compare the experimental probability to the theoretical probability of selecting an odd-numbered card.

2.9.4 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about probability, including those listed below.

- Determine the sample space for a simple experiment
 - Express probabilities as fractions, decimals, and percentages
 - Interpret and compare the probability of chance events
 - Find the theoretical probability of an event related to a simple experiment
 - Compare experimental probability from simulations to theoretical probability
1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck on how to ...

I'm starting to get the hang of ...

I feel confident and ready to go with ...

